

GSOE9210 Engineering Decisions

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Solving games

- 1 Rational play
 - Solutions of zero-sum games
 - Best response
 - Equilibrium
 - Rational solutions; beliefs
- 2 Rational play: non-zero-sum games
 - Cooperation in games

Solving games

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2 Rational play: non-zero-sum games

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Two-player zero-sum games: dominance

Example: simplify/reduce this two-player zero-sum game:

	b_1	b_2	b_3	b_4
a_1	0	1	7	6
a_2	4	2	3	4
a_3	3	1	0	2
a_4	0	0	7	3

- Round 1
- Round 2
- Round 3
- Round 4

- Common knowledge: a player won't play a dominated strategy; other players know this
- Game reduced (iterated dominance) to single strategy for each player: unique solution: (a_2, b_2)

Rational behaviour and strategic uncertainty

- In games the uncertainty for each player includes the *behaviour* of other players; *i.e.*, which strategy they'll choose
- This uncertainty can be reduced if players have *common knowledge* about the preferences and rationality of other players
- Dominance reduces *strategic uncertainty* about rational behaviour of other players (e.g., rational players will never play dominated strategies)
- General principle about rational behaviour: *best response* . . .

Best response

Re-visit previous zero-sum game:

	b_1	b_2	b_3	b_4
a_1	0	1	7	6
a_2	4	②	3	4
a_3	3	1	0	2
a_4	0	0	7	3

- Play (a_2, b_2) is maximal in its column and minimal in its row
- *i.e.*, if column player plays b_2 , then a_2 gives best possible outcome for row player
- Conversely, if row player plays a_2 , then b_2 gives best possible outcome for column player

Best response: zero-sum games

Definition (Best response)

A player's strategy s^* is a *best response* to another player's strategy s if s^* gives a preference maximal outcome against s .

	b_1	b_2
a_1	$\begin{matrix} 2 \\ * \end{matrix}$	$\begin{matrix} 0 \\ * \end{matrix}$
a_2	$\begin{matrix} 1 \\ * \end{matrix}$	$\begin{matrix} 3 \\ * \end{matrix}$

In a zero-sum game:

- for any strategy of the column player, a best response of the row player is a strategy which maximises the column value (*)
- for any strategy of the row player, a best response of the column player is a strategy which minimises the row value (*)

Best response

	b_1	b_2	b_3	min
a_1	$\begin{matrix} 1 \\ * \end{matrix}$	$\begin{matrix} 2 \\ * \end{matrix}$	$\begin{matrix} 6 \\ * \end{matrix}$	1
a_2	$\begin{matrix} 7 \\ * \end{matrix}$	$\begin{matrix} 3 \\ * \end{matrix}$	$\begin{matrix} 3 \\ * \end{matrix}$	3
a_3	$\begin{matrix} 3 \\ * \end{matrix}$	$\begin{matrix} 2 \\ * \end{matrix}$	$\begin{matrix} 5 \\ * \end{matrix}$	2
max	7	3	6	

- Strictly dominated strategies are never best responses to any other player's strategies
- Column player's best responses are minimal in their row
- Every strategy has at least one best response; a_2 has two
- Row player's best responses are maximal in their column
- Multiple best responses must have same payoff

Best response: *Maximin*

	b_1	b_2	b_3	min
a_1	1*	2	6*	1
a_2	7*	3*	3*	3
a_3	3	2*	5	2
max	7	3	6	

- Row player's *Maximin* strategy is best strategy against 'perfect play' by opponent
- Above, row player's *Maximin* strategy is a_2 ; column player's *Maximin* strategy (i.e., *miniMax* strategy) is b_2
- *Maximin* is rational sometimes: e.g., if opponent can see your move

Repeated play

	b_1	b_2	b_3
a_1	1*	2	6*
a_2	4	3*	4
a_3	7*	2*	5

- Suppose initially row player plays a_3 , hoping for best outcome; similarly column player plays b_1 ; play (a_3, b_1)
- Row player happy (best response)
- Column player unhappy, so switches to best response b_2 ; in response row player plays a_2 ; ...
- Play 'stabilises' at (a_2, b_2)

Equilibrium

In 'stable' play (a_2, b_2) each strategy is a best response to the others.

	b_1	b_2	b_3
a_1	1*	2	6*
a_2	4	3*	4
a_3	7*	2*	5



John F. Nash (1928–2015[†])

Definition (Nash equilibrium)

A play is in *equilibrium* if each of its strategies is a best response to the others.

Equilibrium: belief interpretation

	b_1	b_2	b_3
a_1	1*	2	6
a_2	4*	3*	4
a_3	7*	2*	5

- If row player believes column player will play b_2 , then row player cannot improve outcome by switching, and vice versa
- More generally, if each player believes the other will play according to their equilibrium strategy, then neither can improve their outcome by deviating from their equilibrium strategy

Equilibrium: existence and uniqueness

- Not all games have an equilibrium ... in pure strategies

	b_1	b_2
a_1	$\ast 2$	$0^\ast$
a_2	$1^\ast$	$\ast 3$

- Some games have multiple equilibria:

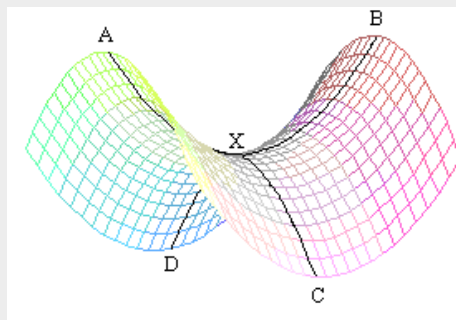
	b_1	b_2	b_3	b_4
a_1	4	$\ast 2^\ast$	5	$\ast 2^\ast$
a_2	2	1	-1	-2
a_3	3	$\ast 2^\ast$	4	$\ast 2^\ast$
a_4	-1	0	6	1

Zero-sum games: finding equilibria

Definition (Saddle point)

An entry in a matrix is called a *saddle point* iff it is minimal in its row and maximal in its column.

	b_1	b_2	b_3
a_1	$1^\ast$	3	4
a_2	$\ast 7$	$\ast 5^\ast$	$6^\ast$
a_3	$3^\ast$	4	$\ast 8$



Theorem (Minimax)

In zero-sum games, saddle points represent equilibria.

Zero-sum games: solutions

Theorem

If a zero-sum game has an equilibrium, then it corresponds to the players playing *Maximin* strategies.

	b_1	b_2	b_3	min
a_1	1	3	4	1
a_2	7	5^*	6	5
a_3	3	4	8	3
max	7	5	8	

Because matrix entries are payoffs for row player, the column player's *Maximin* strategy translates to a *miniMax* strategy.

Zero-sum games: equilibrium

	b_1	b_2	b_3	min
a_1	1	3	4	1
a_2	7	5^*	6	5
a_3	3	4	8	3
max	7	5	8	

Theorem (Unique value)

All equilibria in a zero-sum game yield the same payoff. This payoff is said to be the value of the game.

- The value of the game above is 5
- Equilibria in zero-sum games are paired *Maximin* strategies (*miniMax* for column player)

Zero-sum games: finding equilibria

	b_1	b_2	b_3	min
a_1	1	3	4	1
a_2	7	5^*	6	5
a_3	3	4	8	3
max	7	5	8	

- Saddle points are equilibria in zero-sum games
- To find equilibria:
 - Use *Maximin* to evaluate each of the players' strategies (i.e., *miniMax* for column player)
 - If the *Maximin* values agree (e.g., 5 above), then that play is a saddle point of the game

Equilibrium: existence and uniqueness

- Not all games have an equilibrium ... in pure strategies

	b_1	b_2
a_1	2^*	0
a_2	1	3^*

- Some games have multiple equilibria:

	b_1	b_2	b_3	b_4
a_1	4	2^*	5	2^*
a_2	2	1	-1	-2
a_3	3	2^*	4	2^*
a_4	-1	0	6	1

Behaviour and beliefs

- A game matrix includes all possible strategies and outcomes, but does not specify the players' *behaviour*; *i.e.*, which strategies the players should play
- Dominance and best response are principles about rational *behaviour*
- An agent's behaviour should depend on its preferences and *beliefs* about the other players' behaviour (including likelihoods)
- How do we represent beliefs about other agents' behaviour?

Beliefs and behaviour

- Beliefs about the other players' play can be represented by a mixture of the other players' pure strategies
- Player A assigns to player B's strategy b_j a 'proportion' p_j if A's belief in the 'degree of likelihood' that B will play b_j is p_j
- Recall that utilities encode preferences in the presence of uncertainty

Best response to beliefs: zero-sum games

		B	
		b_1	b_2
A	a_1	2	0
	a_2	1	3

- For belief β calculate the Bayes values of A's strategies:

$$V_B^\beta(a_1) = \frac{1}{2}(2) + \frac{1}{2}(0) = 1$$

$$V_B^\beta(a_2) = \frac{1}{2}(1) + \frac{1}{2}(3) = \frac{4}{2} = 2$$

- Therefore, A's best response given belief β about B is a_2 .
- Player A believes that player B is equally likely to play b_2 as b_1 ; i.e., B will play b_1 with probability $\frac{1}{2}$ and b_2 with probability $\frac{1}{2}$
- Let $\beta \sim (\frac{1}{2}, \frac{1}{2})$ represent A's 'belief' about B's behaviour

Rationalisation of behaviour and belief

- Any strategy by another player which will not be played should receive degree of belief (i.e., probability) 0
- In general, a strategy which isn't Bayes for some belief β can be eliminated; compare with *admissibility*
- In a zero-sum game, rational strategies (strategies which aren't eliminated) must be on the player's 'admissibility frontier'

Solving games

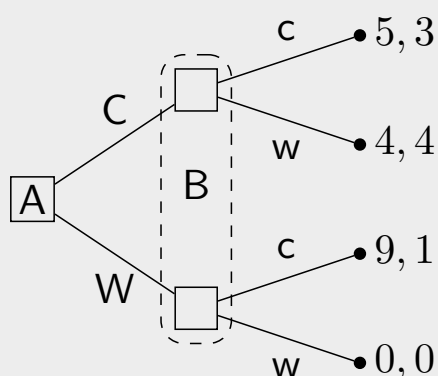
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Non-zero-sum games: best response

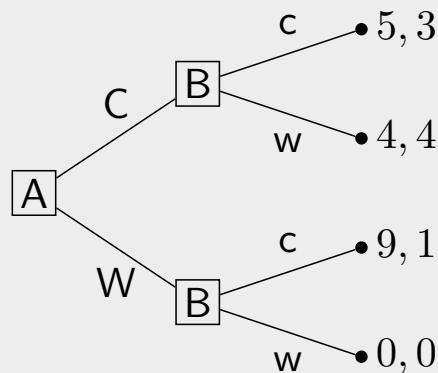


		B	
		c	w
A	C	5, 3	4, 4*
	W	*9, 1	0, 0

- If Alice were to wait, then Bob's best counter-move would be to climb
- Conversely, if Bob were to climb, then Alice's best counter-move would be to wait below

Solving games

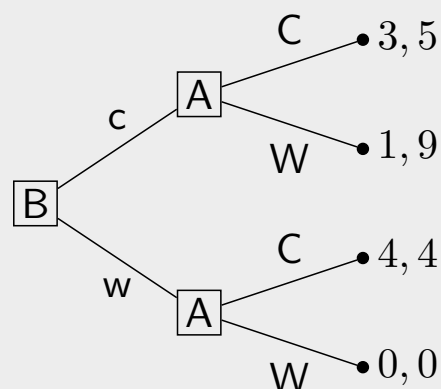
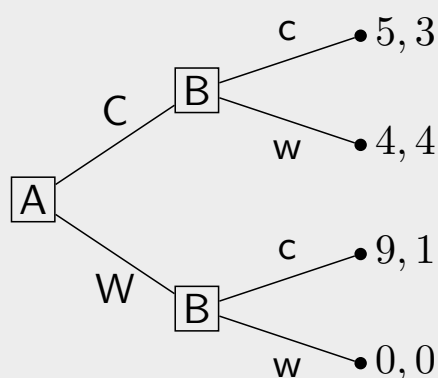
- What if Alice moves first?



Exercises

- What is Bob's best response to Alice waiting? To Alice Climbing?
- Are there any equilibria? If so, which are they?

Equilibrium and solutions



Exercise

For the problems above, find all the equilibrium plays.

- In games that aren't strictly competitive, solutions are less clear because opportunities for co-operation arise
- Other considerations include: group benefit (Pareto optimality), initial tendencies (equilibrium), etc.

Non-strictly-competitive games

Example (The Prisoner's Dilemma)

Alice and Bob are suspects in a joint crime. The police doesn't yet have enough evidence to convict both/either, so it is trying to get either to implicate the other. The police inspector offers each an 'incentive' to 'defect' (D) by implicating their accomplice:

If both suspects defect they will each get a moderate sentence (2 years). If only one defects, the defector will get immunity (0) and the other the full sentence (3 years). If neither defects—*i.e.*, they both cooperate (C)—both will be charged for only a minor offence (1 year).

	d	c
D	1, 1	3, 0
C	0, 3	2, 2

The payoff is the *reduction* in the player's sentence: $3 - s$, where $s \in \{0, 1, 2, 3\}$ is the length of the sentence.

Cooperation in non-zero-sum games

		B	
		d	c
A	D	1, 1	3, 0
	C	0, 3	2, 2

- Individual rationality (dominance) suggests both should defect (Dd); but mutual cooperation (Cc) better outcome for both
- In games which aren't strictly competitive cooperation may be possible
- What's best individually (individual rationality) may not best collectively, and vice versa
- Here Cc gives each player a better payoff than the individually rational play Dd

The Prisoner's Dilemma

Definition (Pareto optimality)

An outcome is *Pareto optimal* iff there is no other outcome which is at least as good or better for all the agents.

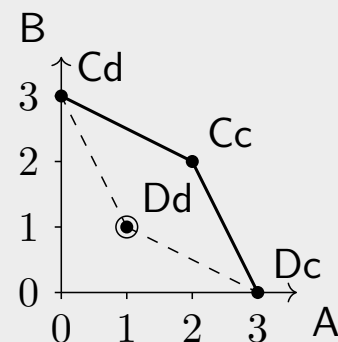
Pareto principle

Pareto optimal outcomes are optimal for a group.

Two-player *play diagram*:

- x -value (abscissa) is A's payoff
- y -value (ordinate) is B's payoff

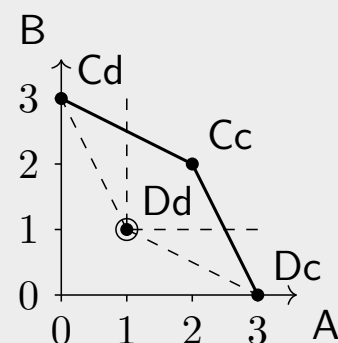
Pareto optimal outcomes represented by points on solid line



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The Prisoner's Dilemma

		B	
		d	c
A	D	1, 1	3, 0
	C	0, 3	2, 2

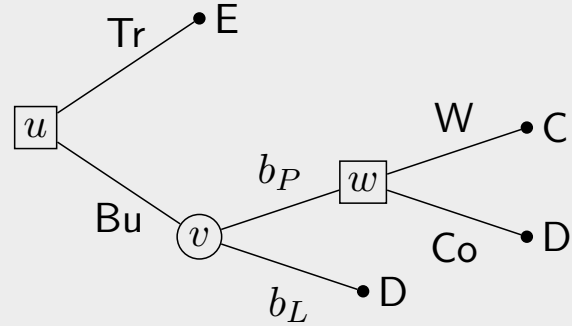
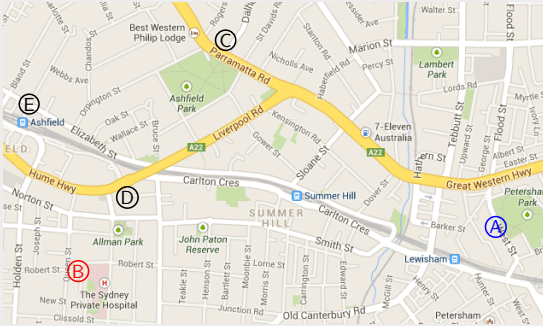


- The equilibrium is Dd (circled)
- The Pareto optimal outcomes are: Cc, Cd, Dc
- Play Cc, which is Pareto optimal, is better than Dd for both players

Conclusion

In two-player non-strictly-competitive games, what's best for the individual may not be best for the group; *i.e.*, *cooperation* desirable.

'Nature' as a player



- Can regarded single-agent decisions as games against a neutral player called 'Nature', or 'Chance', who has no preferences*
- Games in which some of the players' preferences (and/or playing history) are unknown are said to have *incomplete information*—as opposed to *imperfect information*, in which information sets may have multiple nodes
- In extensive form, Nature's moves take place at chance nodes, and they correspond to chance events

Summary

- Best response strategies
- Equilibrium in zero-sum games: saddle-points
- Solutions: rational reduction
- Non-zero-sum games: group preference and Pareto optimality; cooperation
- Single agent decisions are 'games against nature'